Engram

Engram checks a medical recall against your stock — and shows you the proof for every answer.

Open Agent Hackathon 2026 · Track 2: The Agent That Can Explain Why

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BUILT ON

ZETARIS

Meterless Nightshift

NVIDIA Nemotron 3.5 Super

In a real 2025 recall, two official documents disagree by one character.

The distributor letter says

23IJA103

≠

The FDA and the manufacturer say

23JJA103

FDA recall Z-1706-2025: Medline 100 mL saline with leaking bottle seals, 32 affected lots. Match the wrong code and recalled bottles stay on the shelf — or safe stock gets thrown away.

Today, a hospital recall coordinator does all of this by hand, for every notice.



“Repeated notice review, fragmented communication and local operational work.”

How practitioners from Mayo, Yale New Haven and Froedtert describe it — Healthcare Value Analysis Magazine, “Speed to Safety” roundtable.

Who it’s for: recall coordinators, hospital pharmacy, materials and value-analysis teams.

You ask one question. Engram answers with evidence and a plan you can check.

YOU ASK

"Which stock is affected by this recall, and can every site still cover its demand?"

ENGRAM ANSWERS

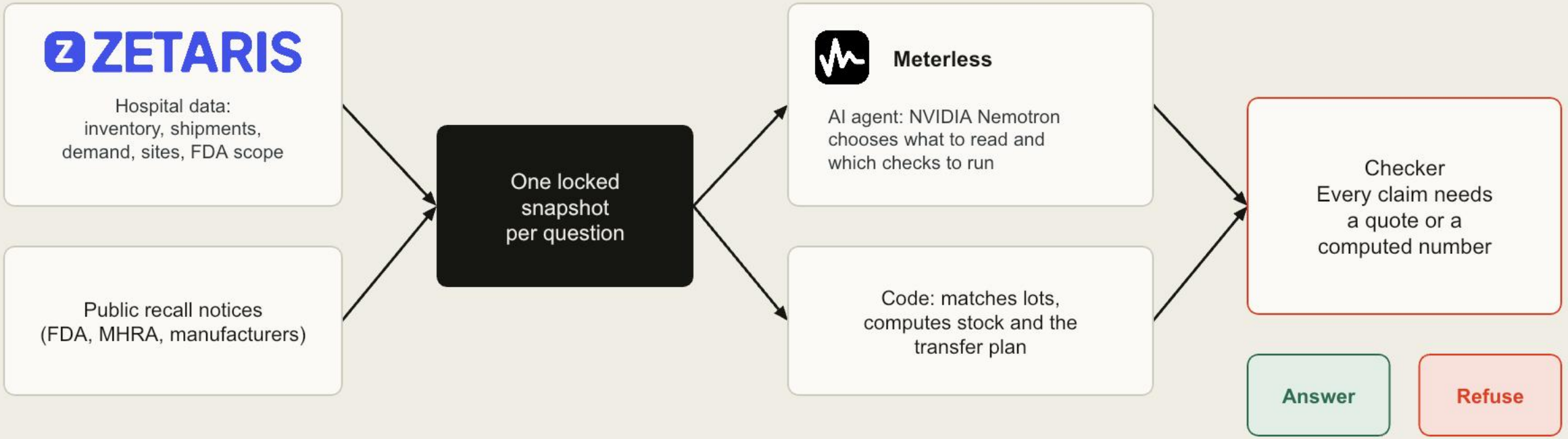
120 bottles are taken out of supply — their lots match the FDA list

10 bottles wait for a human — their lot code is missing

30 bottles move from site B to site A — every site stays above its safety reserve

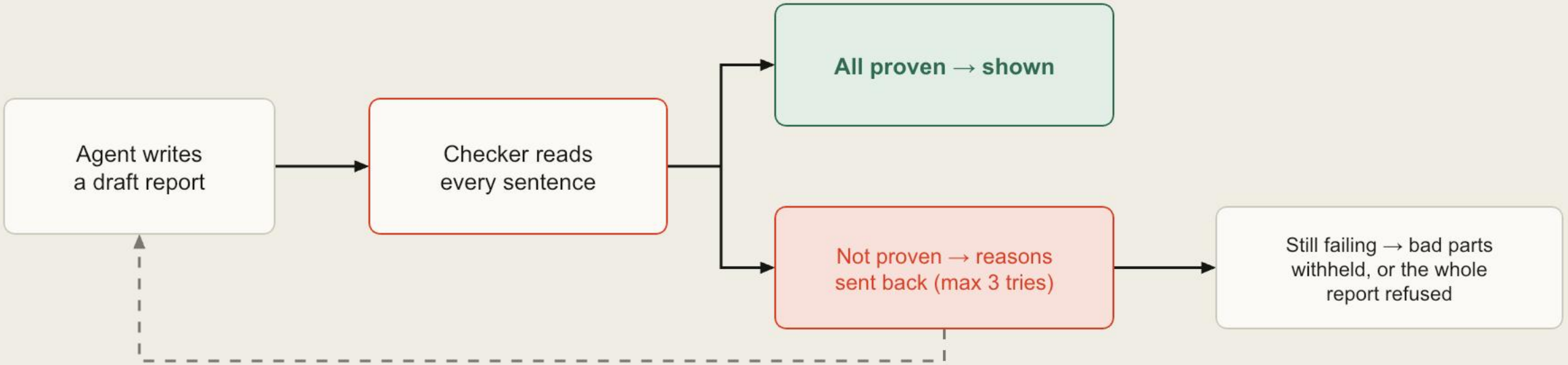
Every number links back to the exact source row or notice text. Nothing is physically moved — approvals are recorded in a sandbox only.

The agent reads and decides what to check. Code does the maths. A checker signs off.



Engram shows the evidence and says “human review required” when a claim can’t be proven.

If a sentence can’t be traced to a source, Engram won’t show it.



WHAT IT HAS CAUGHT (paraphrased from real rejected drafts)

A number with no source

“23 batches” when the table lists 22

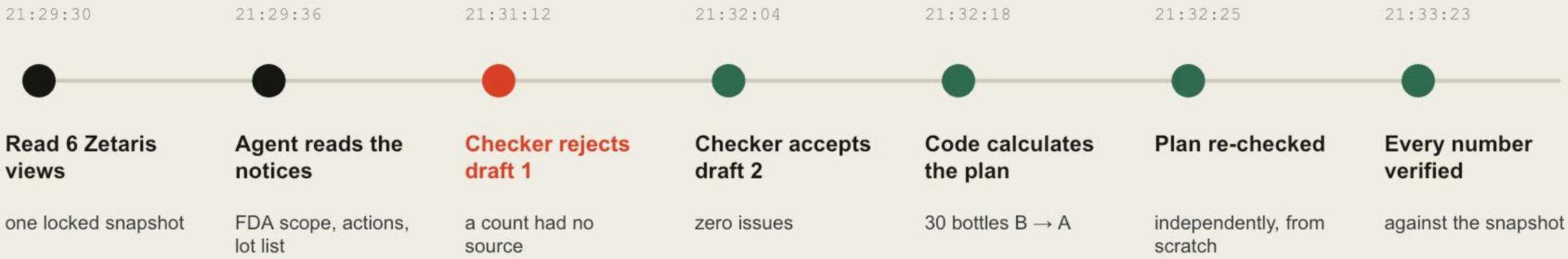
A change the notice never made

“the new notice narrows the recall”

A claim about stock the model can’t see

“these records are safe”

Tonight's live run, step by step: 3 minutes 53 seconds from question to checked plan.



Live on a laptop: Zetaris → Meterless → Nemotron (11 model calls) → checked plan.
Hospital data is fictional; the recall is real.
In the same run, Meterless H-MEM saved this investigation to memory for the next one.

Where it works today, and where it refuses.

Live stock check (Medline saline, FDA)	**Answered — every number verified**
UK notice update (ChloraPrep, MHRA)	**Partly answered — 3 findings shown, 2 withheld**
Device notice update (Olympus ViziShot)	**Refused — a claim about stock it couldn't prove**
Pump notice update (Medtronic MiniMed)	**Refused — said "narrows the recall" without proof**

A refusal is the safe failure: you see the evidence and "human review required", never a confident wrong answer. The three notice cases were used while building, so they are not an independent test.

Real recall notices. A made-up hospital. Nothing about patients.

REAL — public documents

FDA recall Z-1706-2025 — Medline saline, 32 lots

MHRA ChloraPrep recalls — May and July 2025

Olympus ViziShot letters — 2025 and 2026

Medtronic MiniMed notices — 2019 and 2021

MADE UP — served through Zetaris

3 sites with stock, demand and safety reserves

Shipments, routes and QA records

Earlier decisions to re-check

Each question locks its own data snapshot

Prototype — not a medical device and not a regulatory tool.

Zetaris holds the hospital data. Meterless runs the AI model and remembers past reviews.

ZETARIS

The hospital's data source. Engram reads 6 SQL views through the Zetaris REST API and locks them as one snapshot, so every number in an answer traces back to a row.

Tonight: 6 views in 34.6 s.

Meterless

Nightshift is the only way Engram reaches the model; the model route is verified on every reply.

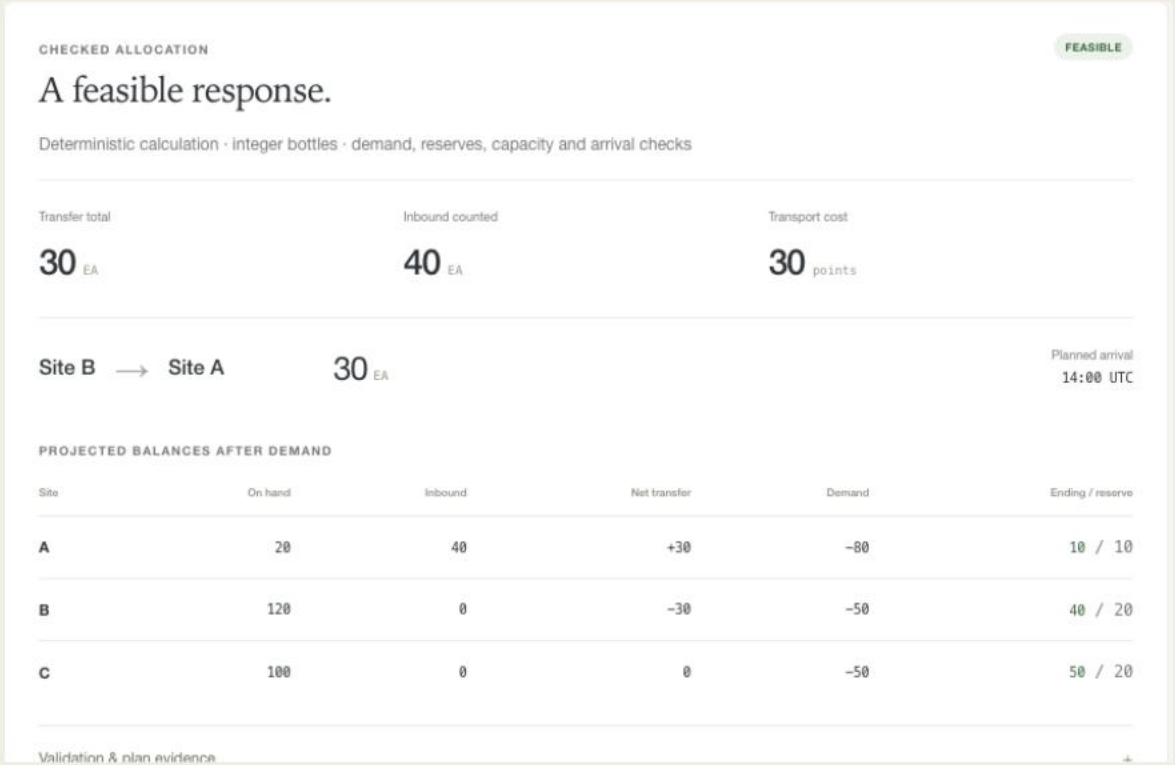
H-MEM memory engine: reviewer decisions are recalled before each investigation, and each run is stored for the next (used live tonight; covered by tests).

NVIDIA

Nemotron 3.5 Super runs every agent step — choosing sources, calling tools and writing the report the checker reviews.

Built with Claude Code and Codex. Cursor is optional per the organisers' HackOS announcement of 22 October.

What a recall coordinator sees: the plan, and the checked claims behind it.



CHECKED ALLOCATION

FEASIBLE

A feasible response.

Deterministic calculation · integer bottles · demand, reserves, capacity and arrival checks

Transfer total	Inbound counted	Transport cost
30 EA	40 EA	30 points

Site B → Site A 30 EA

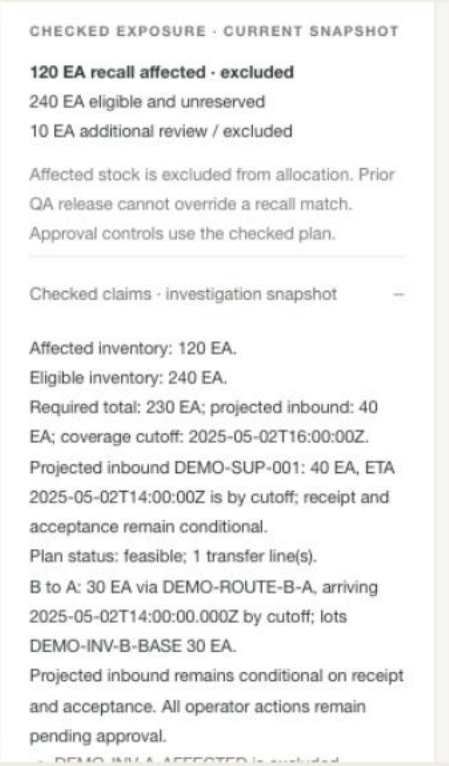
Planned arrival
14:00 UTC

PROJECTED BALANCES AFTER DEMAND

Site	On hand	Inbound	Net transfer	Demand	Ending / reserve
A	20	40	+30	−80	10 / 10
B	120	0	−30	−50	40 / 20
C	100	0	0	−50	50 / 20

Validation & plan evidence +

The checked plan: who sends how many bottles, and what each site has left after demand.



CHECKED EXPOSURE · CURRENT SNAPSHOT

120 EA recall affected · excluded
240 EA eligible and unreserved
10 EA additional review / excluded

Affected stock is excluded from allocation. Prior QA release cannot override a recall match. Approval controls use the checked plan.

Checked claims · investigation snapshot −

Affected inventory: 120 EA.
Eligible inventory: 240 EA.
Required total: 230 EA; projected inbound: 40 EA; coverage cutoff: 2025-05-02T16:00:00Z.
Projected inbound DEMO-SUP-001: 40 EA, ETA 2025-05-02T14:00:00Z is by cutoff; receipt and acceptance remain conditional.
Plan status: feasible; 1 transfer line(s).
B to A: 30 EA via DEMO-ROUTE-B-A, arriving 2025-05-02T14:00:00.000Z by cutoff; lots DEMO-INV-B-BASE 30 EA.
Projected inbound remains conditional on receipt and acceptance. All operator actions remain pending approval.

Verified claims, each linked to its source.

What's next

- Test on recall notices it has never seen
- Catch "offer" vs "requirement" wording errors
- Check device details, not only lot and product
- Pilot with real recall coordinators

A recall assistant that shows its proof — and says so when it has none.